

End-to-End Quadrotor Position Control via Proximal Policy Optimization with Direct Motor Actuation

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Abstract

Reinforcement learning (RL) offers a path to fully autonomous quadrotor control by mapping raw sensor observations directly to motor commands without hand-engineered control hierarchy. In this paper we present a complete end-to-end learning pipeline for quadrotor position control at the direct motor RPM level using Proximal Policy Optimization (PPO). Training within a physics-accurate Crazyflie 2.x simulation built on the Newton GPU rigid-body engine, our policy achieves 82.9% per-leg waypoint success after 30 million environment steps under clean-start conditions (81.8% at 40M). We characterise learning progression from 3M to 40M steps under both clean-start and fully randomised spawn conditions, provide distance-stratified success rates, and quantify a robustness gap that narrows consistently over training (47.1 pp at 3M $\rightarrow$ 28.3 pp at 40M).

Keywords: Reinforcement learning, Proximal Policy Optimization, Direct motor control, Curriculum learning

I. Introduction

Classical quadrotor control stacks independently tuned layers — trajectory planner, position controller, attitude stabiliser, and motor mixer — each requiring explicit dynamic models and brittle to unmodelled dynamics. RL offers a single neural network mapping raw sensor readings directly to motor commands, eliminating per-layer tuning. At the direct RPM level the policy must learn all control layers simultaneously from one reward signal, facing quadratic thrust ($F = K_T \cdot n^2$), motor lag ($\tau \approx 0.15$ s), and the full inertia tensor — making reward design the central challenge. PPO [1] places especially strong demands on reward density since rollout data is discarded after each update.

Curriculum learning [2] gradually increases training difficulty. We scale spawn randomisation — position offset, velocity, tilt — by a curriculum coefficient $c \in [0, 1]$ that ramps over the first 1.5 M steps. Reward weights are simultaneously annealed, allowing the policy to learn hovering before being challenged by recovery from disturbed states.

Our contributions: (1) A reward function with coupled potential-based approach and velocity-gated settlement terms, both jointly necessary for convergence. (2) Training progression characterised from 3M to 40M steps under clean-start and randomised spawn conditions. (3) Distance-stratified success rates and a spawn robustness comparison showing a gap that narrows from 47.1 pp at 3M to 28.3 pp at 40M. (4) An open Crazyflie 2.x training environment using Newton GPU physics and Stable-Baselines3 [1].

II. Method

A. Simulation Environment

The environment models a Crazyflie 2.x nano-quadrotor using system-identified parameters [3],

integrated at 100 Hz via Newton's SolverSemiImplicit. A first-order LPF ($\tau = 0.15$ s) models motor lag. During training, sixteen environments run in parallel with each episodes last up to 800 steps (8.0 s). Key parameters are in Table 1.

Table 1. Crazyflie 2.x Physical Parameters

Parameter	Value
Mass	27 g
Arm length	32.5 mm
$I_{xx} = I_{yy}$	1.657×10^{-5} kg·m ²
I_{zz}	2.900×10^{-5} kg·m ²
K_T	3.16×10^{-10} N/RPM ²
K_D	7.94×10^{-12} N·m/RPM ²
Hover RPM	$\approx 14,476$ RPM

B. Observation and Action Space

The 22D observation encodes: position error $p_{err} = pos - target$ (3D), rotation matrix R row-major (9D, avoids quaternion double-cover), linear velocity v (3D, world frame), angular velocity ω (3D, body frame), previous action a_{prev} (4D). The 4D action maps normalised setpoints $\in [-1, 1]$ to motor RPM: $n_{sp,i} = clip(14476 + a_i \cdot 7226, 1000, 21702)$ RPM. Action 0 = hover equilibrium.

C. PPO Configuration

Stable-Baselines3 PPO [4]: shared-trunk MLP (two 256-unit layers, tanh), separate actor and critic heads. Rollout length 2,048 steps/env (32,768 transitions/update), exceeding the 800-step episode limit so GAE returns are never truncated mid-episode. $n_{epochs} = 10$, $batch_size = 64$, $\gamma = 0.99$, $\lambda = 0.95$, $clip = 0.2$, $ent_coef = 0.005$.

D. Reward Function

The per-step reward sums ten components shown in Table 2. The gated approach term suppresses oscillation incentives inside $d_{\text{gate}} = 0.10$ m while maintaining dense transit gradient right up to the 0.15 m success boundary. The velocity-gated hover bonus — active only when $\|v\| < 0.5$ m/s — is critical: without the speed condition, the policy learns to sprint at the target to enter the bonus zone sooner, causing frequent overshoots.

Table 2. Reward Function Components

Term	Formula	Role
pos_c	$-C_{rp} \cdot \ p_{\text{err}}\ ^2$	Position precision
orient_c	$-C_{rq} \cdot (1 - q_w^2)$	Upright attitude
vel_c	$-C_{rv} \cdot \ v\ ^2$	Velocity damping
ang_c	$-C_{r\omega} \cdot \ \omega\ ^2$	Angular stability
act_c	$-C_{ra} \cdot \ \Delta a\ ^2$	Action jerk
survival	+ 0.50 (fixed)	Stay airborne
approach	$+ \lambda \cdot (d_{\{t-1\}} - d_t), d > 0.10$ m	Dense transit signal
hover_bonus	+ 0.15, $d < 0.15$ m & $\ v\ < 0.5$ m/s	Settlement reward
arrival	+ 15.0 (one-time)	Waypoint reached
crash	- 2.0 (one-time)	Ground impact

Weights ramp over first 1.5 M steps: C_{rp} 0.05→1.00, C_{rv} 0.01→0.30, $C_{r\omega}$ 0.001→0.05, C_{ra} 0.005→0.02. $C_{rq} = 0.10$ fixed.

E. Spawn Curriculum

Each episode the drone spawns near the first waypoint with offsets scaled by c : position $\leq (0.1+0.6c)$ m, velocity $\leq 1.5c$ m/s, angular rate $\leq 0.5c$ rad/s, tilt $\leq 15c^\circ$. At $c = 0$ (clean start): stationary, upright, ≤ 0.1 m from waypoint. At $c = 1.0$ (full randomisation): significant velocity and tilt requiring attitude recovery during navigation.

F. Experiment

Evaluation uses multi-waypoint chaining (sequential waypoints, no reset, 8,000-step budget, 20 episodes). Clean-start ($c=0$): stationary, upright spawn. Randomised ($c=1.0$): full curriculum distribution. A leg succeeds when $\|p_{\text{err}}\| < 0.15$ m. Table 2 reports performance at three checkpoints. Under clean-start, convergence occurs at ~ 30 M (+9.9 pp, then -1.1 pp). Randomised performance continues improving through 40M (25.9%→53.5%), with the robustness gap narrowing monotonically (47.1→40.0→28.3 pp), confirming robustness is a slowly-learned property that outlasts clean-start convergence.

Table 3. Learning Progression by Spawn Mode

Ckpt	Cl. SR	Cl. Wps	Rd. SR	Rd. Wps	Gap
3M	73.0%	2.7±2.3	25.9%	0.3±0.7	47.1 pp
30M	82.9%	4.8±3.2	42.9%	0.8±0.8	40.0 pp
40M	81.8%	4.5±3.6	53.5%	1.1±1.8	28.3 pp

SR: leg success rate (fraction of waypoint legs reached within 0.15 m). Wps: mean waypoints reached per episode $\pm$ std. Cl.: clean-start ($c=0$). Rd.: randomised ($c=1.0$).

III. Conclusion

We presented a complete end-to-end learning pipeline for direct motor RPM control of a nano-quadrotor using PPO. Training to 40M steps achieves 81.8% per-leg waypoint success under clean-start conditions (convergence at ~ 30 M). Distance-stratified results show degradation from 100% at < 0.5 m to 68.2% at 1.5–3.0 m. A key finding is that robustness under disturbed spawn conditions continues to improve beyond clean-start convergence — the gap narrows from 47.1 pp at 3M to 28.3 pp at 40M, with randomised performance rising from 25.9% to 53.5%. Future work will target: (1) domain randomisation of motor lag parameters for improved robustness; (2) replacing the flat survival bonus to eliminate the hover attractor; and (3) zero-shot Sim-to-Real validation on a physical Crazyflie 2.x.

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REFERENCES

- [1] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- [2] Bengio, Y., Louradour, J., Collobert, R., & Weston, J. (2009, June). Curriculum learning. In *Proceedings of the 26th annual international conference on machine learning* (pp. 41–48).
- [3] D. Förster, "System Identification of the Crazyflie 2.0," B.Sc. thesis, ETH Zürich, 2015.
- [4] Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., & Dormann, N. (2021). Stable-baselines3: Reliable reinforcement learning implementations. *Journal of machine learning research*, 22(268), 1–8.